

Def. Let V be a vector space.

Given a vector $\alpha \in V$ and linear operator $T: V \rightarrow V$, the subspace

$$Z(\alpha, T) \stackrel{\text{def}}{=} \{g(T)\alpha \mid g(t) \in F[t]\} = \text{span}\{T^k \alpha \mid k \in \mathbb{N} \cup \{0\}\}.$$

is called the T -cyclic subspace generated by α

Def. Let V be a vector space. Given $\alpha \in V$ and a linear operator T on V , an ideal

$$M(\alpha, T) = \{g(t) \in F[t] \mid g(T)\alpha = 0\}$$

is called the T -annihilator of α , and the unique monic generator is also called μ .

Thm. Let α be any non-zero vector in V and let p_α be the T -annihilator of α .

1) $\deg p_\alpha = \dim Z(\alpha, T)$

2) $Z(\alpha, T) = \text{span}\{T^k \alpha \mid 0 \leq k \leq \deg p_\alpha\}$

3) The minimal polynomial of $T|_{Z(\alpha, T)}$ is p_α .

Proof. Given a polynomial $g \in F[x]$, the Euclidean Algorithm gives

$$g = p_\alpha \cdot q + r \quad \text{for some } q, r \in F[x], \text{ with } r = 0 \text{ or } \deg r < \deg p_\alpha.$$

Then, $g(T)\alpha = p_\alpha(T)g(T)\alpha + r(T)\alpha = r(T)\alpha$. Since either $r = 0$ or $\deg r < \deg p_\alpha$, the vector $r(T)\alpha$ is a linear combination of $\{\alpha, T\alpha, T^2\alpha, \dots, T^{\deg p_\alpha - 1}\alpha\}$.

Since g was arbitrary, $Z(\alpha, T)$ is spanned by $\{T^n \alpha \mid n = 0, 1, \dots, \deg p_\alpha - 1\}$.

Moreover, these are linearly independent, because: if there exists non-trivial representation

$$c_0 \alpha + c_1 T\alpha + \dots + c_{\deg p_\alpha - 1} T^{\deg p_\alpha - 1} \alpha = 0,$$

then it contradicts to the minimality of p_α . It proves 1) and 2).

To show the third assertion, let $U = T|_{Z(\alpha, T)}: Z(\alpha, T) \rightarrow Z(\alpha, T): \alpha \mapsto T\alpha$.

It is well-defined, since the cyclic subspace is invariant under T .

Given a polynomial $g \in F[x]$,

$$p_\alpha(U)g(T)\alpha \underset{g(T)\alpha \in Z(\alpha, T)}{=} p_\alpha(T)g(T)\alpha = g(T)p_\alpha(T)\alpha = g(T)0 = 0.$$

$$g(T)\alpha \in Z(\alpha, T)$$

So, $p_\alpha(U)$ is identically zero. Moreover, $h(U)$ cannot be zero if $\deg h < \deg p_\alpha$, because $h(U)\alpha = h(T)\alpha = 0$, then it contradicts to the minimality.

Companion Matrix.

Let W be a vector space of dimension $k \in \mathbb{N}$, and $T: W \rightarrow W$ be a linear operator which have a cyclic vector $\alpha \in W$. By the Theorem above, $\{\alpha, T\alpha, \dots, T^{k-1}\alpha\}$ forms a basis for W . As above fact and the Cayley-Hamilton Theorem, let

$$p_\alpha(x) = x^k + c_{k-1}x^{k-1} + \dots + c_1x + c_0$$

be the minimal polynomial (and also characteristic) for T .

Define $\beta = \{T^{n-1}\alpha \mid n=1, 2, \dots, k\}$, denote $d_n = T^{n-1}\alpha$. Then,

$$0 = 0 \cdot \alpha = p_\alpha(T)\alpha = T^k\alpha + c_{k-1}T^{k-1}\alpha + \dots + c_1T\alpha + c_0\alpha$$

$$\Rightarrow T^k\alpha = -c_0\alpha - c_1T\alpha - \dots - c_{k-1}T^{k-1}\alpha$$

$$= -c_0d_1 - c_1d_2 - \dots - c_{k-1}d_k$$

$$[T]_\beta = \begin{bmatrix} 0 & 0 & \dots & 0 & -c_0 \\ 1 & 0 & \dots & 0 & -c_1 \\ 0 & 1 & \dots & 0 & -c_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & \dots & 0 & 1 & -c_{k-1} \end{bmatrix} = \begin{bmatrix} 0 & \dots & 0 & -c_0 \\ \vdots & & & \vdots \\ I_{k-1} & & & -c_{k-1} \end{bmatrix}$$

Directly, we obtain that: on a fin. dim space T ,

A linear operator T has a cyclic vector

if and only if

the minimal polynomial of T and characteristic polynomial of T are identical.

Exercises in Hoffman, Sec 7.2

1. Every linear operator T on two-dimensional either:

T has a cyclic vector or T is a scalar multiple of the identity.

Proof. Suppose that T has no cyclic vector. That is, for all $\alpha \in V$,

$\{\alpha, T\alpha\}$ is linearly dependent. Therefore, there is $c \in F$ such that $c\alpha = T\alpha$.

Let $\alpha_1 \in V$ be a non-zero vector. For some $c_1 \in F$, $c_1\alpha_1 = T\alpha_1$.

Now, let $\alpha_2 \in V \setminus \langle \alpha_1 \rangle$. Then, similarly, $c_2\alpha_2 = T\alpha_2$, for some $c_2 \in F$.

By the construction, $\alpha_1 - \alpha_2 \neq 0$, and there is $c_3 \in F$ such that $c_3(\alpha_1 - \alpha_2) = T(\alpha_1 - \alpha_2) = c_1\alpha_1 - c_2\alpha_2$, so

$$(c_3 - c_1)\alpha_1 = (c_3 - c_2)\alpha_2$$

But, since $\alpha_2 \in V \setminus \langle \alpha_1 \rangle$, $c_3 - c_1 = c_3 - c_2 = 0$, that is, $c_1 = c_3 = c_2$.

Now, since $\{\alpha_1, \alpha_2\}$ is basis for V , $T = c_1 I$. $\square$

4. If T^2 has a cyclic vector, then T has a cyclic vector.

Proof. Let $\alpha \in V$ be the cyclic vector of T^2 . That is,

$\mathcal{B} = \{T^{2k}\alpha \mid k=0,1,\dots\}$ spans the space V . Clearly, $\mathcal{B}' = \{T^k\alpha \mid k=0,1,\dots\}$ satisfies $\mathcal{B} \subseteq \mathcal{B}'$, therefore $V = \langle \mathcal{B} \rangle \subseteq \langle \mathcal{B}' \rangle = V$.

But the converse does not hold: let $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $\alpha = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Then, $\{\alpha, A\alpha\} = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}$ spans F^2 but

$A^2 \equiv 0$ has no cyclic vector.

5. Let V be an n -dimensional over F , and let N be a nilpotent on V . Suppose that $N^{n-1} \neq 0$ and let $\alpha \in V$ be any vector such that $N^{n-1}\alpha \neq 0$. Prove that α is a cyclic vector for N .

Proof Since N is nilpotent operator, $N^k = 0$ for some $k \geq 0$, therefore the minimal polynomial of N must be power of x .

But, since $N^{n-1} \neq 0$, so $N, N^2, \dots, N^{n-2}$ are not zero,

by the Cayley-Hamilton Theorem, $N^n = 0$.

For the given $\alpha \in V$, $N^{n-1}\alpha \neq 0$ implies that $\alpha, N\alpha, N^2\alpha, \dots, N^{n-1}\alpha$ are all not zero. (if $N^{n-k}\alpha = 0$, then $N^{k-1} \cdot N^{n-k}\alpha = 0$) Moreover, these are mutually distinct, because: for $0 \leq i < j \leq n-1$, $N^i\alpha = N^j\alpha$ implies $N^{j+n-i}\alpha = N^{i+n-j}\alpha = 0$,

but $i+n-j < n$, it is contradiction. Moreover, these are linearly independent.

Suppose that $C_0\alpha + C_1N\alpha + \dots + C_{n-1}N^{n-1}\alpha = 0$, $C_i \in F$.

Take $N^{n-1}, N^{n-2}, \dots, N$ to the both side, then we obtain

$C_0 = 0, C_1 = 0, \dots, C_{n-1} = 0$. So, the set $\{\alpha, \dots, N^{n-1}\alpha\}$ is linearly independent set of n -elements in the n -dimensional space V , so proved. $\square$

7. Let V be an n -dimensional, and $T: V \rightarrow V$ is diagonalizable linear operator. Then, T has a cyclic vector if and only if T has n -distinct eigenvalues.

Proof. Suppose that T has a cyclic vector $\alpha \in V$.

Since T is diagonalizable, there are linearly independent n -vectors $\alpha_1, \dots, \alpha_n$ such that $\alpha = \alpha_1 + \dots + \alpha_n$ and $T\alpha_i = C_i \alpha_i$ for some $C_i \in F$.

Observe that for each $k \geq 0$,

$$T^k \alpha = T^k (\alpha_1 + \dots + \alpha_n) = \sum_{i=1}^n C_i^k \alpha_i.$$

These are linearly independent if and only if

$\begin{bmatrix} 1 & C_1 & \dots & C_1^{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & C_n & \dots & C_n^{n-1} \end{bmatrix} x = 0$ has only trivial solution, so the matrix is invertible. It implies $C_1, \dots, C_n$ are distinct.

Conversely, let $\beta = \{\alpha_1, \dots, \alpha_n\}$ be a basis of eigenvectors for T corresponding to distinct scalar values $C_1, \dots, C_n$. Then, $\alpha = \alpha_1 + \dots + \alpha_n$ is a cyclic vector, because it is enough to show that $\{\alpha, T\alpha, \dots, T^{n-1}\alpha\}$ is linearly independent.

Suppose that

$$\begin{aligned} x_0 \alpha + x_1 T\alpha + \dots + x_{n-1} T^{n-1} \alpha &= (x_0 + x_1 C_1 + \dots + x_{n-1} C_1^{n-1}) \alpha_1 \\ &\quad + (x_0 + x_1 C_2 + \dots + x_{n-1} C_2^{n-1}) \alpha_2 \\ &\quad \vdots \\ &\quad + (x_0 + x_1 C_n + \dots + x_{n-1} C_n^{n-1}) \alpha_n = 0. \end{aligned}$$

Since $\alpha_1, \dots, \alpha_n$ are linearly independent, therefore

$$x_0 + x_1 C_1 + \dots + x_{n-1} C_1^{n-1} = \dots = x_0 + x_1 C_n + \dots + x_{n-1} C_n^{n-1} = 0.$$

It equals to $\begin{bmatrix} 1 & C_1 & \dots & C_1^{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & C_n & \dots & C_n^{n-1} \end{bmatrix} \begin{bmatrix} x_0 \\ \vdots \\ x_{n-1} \end{bmatrix} = 0$. The matrix is called Vandermonde, it is invertible when $C_1, \dots, C_n$ are distinct. Therefore, it has only trivial solution, $x_0 = \dots = x_{n-1} = 0$. So $\{\alpha, \dots, T^{n-1}\alpha\}$ is linearly independent.

8. Let T be a linear operator on the finite dimensional space V . If T is cyclic and T commute with a linear operator U , then U is a polynomial in T .

Proof. We want to show that: for some polynomial $g(t) \in F[t]$,

$$U\beta = g(T)\beta \quad \text{for all } \beta \in V.$$

Let $\alpha \in V$ be the cyclic vector of T . Then, $\{\alpha, T\alpha, \dots, T^{n-1}\alpha\}$ forms basis for V . Therefore, for some scalars $C_0, \dots, C_{n-1}$,

$$U\alpha = C_0\alpha + C_1T\alpha + \dots + C_{n-1}T^{n-1}\alpha.$$

Now, put $g(t) = C_0 + C_1t + \dots + C_{n-1}t^{n-1}$. Then

$$U\alpha = g(T)\alpha,$$

$$UT\alpha = T U\alpha = T g(T)\alpha = g(T)T\alpha$$

$$\vdots$$

$$UT^{n-1}\alpha = g(T)T^{n-1}\alpha$$

Thus, $U = g(T)$. $\square$